

Ravi Prakash Srivastava

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Education

Indian Institute of Information Technology Ranchi (IIIT Ranchi), India M.Tech (Data Science & AI) <i>Institute of National Importance, Govt. of India</i>	August 2024 – April 2026 CGPA: 8.45/10 (ongoing)
North Eastern Hill University (NEHU), Shillong, India B.Tech (Information Technology) <i>Central University, Govt. of India</i>	August 2018 – June 2022 CGPA: 7.85/10

Experience

Software Engineer SS Software Solutions LLC, Hyderabad, India	March 2026 – Present
- Defined and enforced ML pipeline standards including versioning, logging, and monitoring to improve system reliability and debuggability. Currently learning and applying Threat Modeling to proactively identify design-level risks. - Automated security testing under DevSecOps by integrating tools into CI/CD pipelines, reducing manual effort and improving efficiency.	
Research Data Scientist Pantech e-learning, Chennai, India	February 2023 – September 2024
- Designed and deployed ML-based intrusion and anomaly detection models on real-world datasets (10K+ samples), improving threat detection accuracy by 15–20% under noisy and incomplete data conditions. - Conducted web application penetration testing and vulnerability assessments, identifying critical weaknesses.	
Machine Learning Intern C-DAC (Govt. of India), Hyderabad, India	August 2022 – February 2023
- Performed large-scale data preprocessing and feature engineering on student activity logs (attendance, assignments, quizzes) to support predictive modeling. - Implemented and benchmarked multiple machine learning models, including Decision Trees, Random Forests, and Support Vector Machines, for academic performance prediction.	

Publications

- Adversarial Prompt Injection Attacks on LLM: Cryptographic Key Leakage and Defense Strategies**
Under Publication at International Conference on Advanced Computing and Applications (ICACA-2026).
- IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions using Deep Learning**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2025.2521688>
- Deepfake Video Detection Using Face-Centric Processing and Frame Sampling**
Under Publication International Conference on Advanced Computing and Applications (ICACA-2026).
- Balanced Few-Shot Episodic Learning for Accurate Retinal Disease Diagnosis**
(arXiv: <https://arxiv.org/abs/2512.04967v1>)
- Smart City Vehicle Accident Monitoring and Detection System Using MEMS, GSM, GPS Raspberry Pi 4**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2022.2043787>

Projects

Adversarial Prompt Injection Attacks on LLMs: Cryptographic Key Leakage and Defense Strategies:

- Conducted security analysis of LLM systems by designing prompt injection attack scenarios to exploit context leakage and unauthorized data exposure.
- Developed and evaluated mitigation strategies including prompt sanitization, input/output filtering, and instruction hierarchy enforcement to defend against injection-based attacks.

IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions Using Deep Learning:

- Developed an IoT-based border security system using deep learning for real-time human intrusion detection, aligned with threat modeling (STRIDE) principles.
- Designed system architecture, integrated sensors, and implemented secure detection pipelines with alert mechanisms, incorporating basic security testing and risk mitigation.

Technical Skills

- **AI:** Deep Learning, Transformer Models, Adversarial Machine Learning, Prompt Injection Attacks, Deepfake Detection, Representation Learning, Model Evaluation, Feature Engineering
- **Cybersecurity:** Threat Modeling, Vulnerability Assessment, Web Security (OWASP Top 10), Network Security, Burp Suite, Nmap, Wireshark, Metasploit (Basics)
- **Security Automation & DevSecOps:** Vulnerability Scanning, CI/CD Security, Threat Detection Pipelines
- **Tools & Frameworks:** PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas
- **Programming:** Python, C++, SQL, Bash
- **Platforms & Tools:** Linux, Git, Jupyter Notebook, LaTeX

Achievement

- Received the Institution of Electronics and Telecommunication Engineers (IETE) Scholarship for academic excellence during M.Tech studies in Computer Science and Engineering.
- Best Paper Award, ICACA 2026 for the paper *“Adversarial Prompt Injection Attacks on Large Language Models: Cryptographic Key Leakage and Defense Strategies”*.

Professional Affiliations

- The Institution of Electronics and Telecommunication Engineers (IETE) – Associate Member [AM-503427]